Chemistry • States, mixtures and solutions

States of Matter and Phase Changes

Official exercise • Complete solved answers • Original concept diagrams

Chapter 1 • Complete official exercise • 10 MCQs + 8 short + 8 constructed + 6 descriptive + 2 investigative

What this chapter is about

• **Chemistry** studies the composition, structure, properties and changes of matter, together with the energy and principles associated with those changes.
• **Matter** has mass and occupies space. Particle spacing, motion and attraction explain the properties of solids, liquids and gases; plasma and intermediate or exotic states extend this model.
• **Elements and compounds** are pure substances, whereas mixtures contain components in variable proportions. Solutions, colloids and suspensions differ in particle size and behaviour.
• **Solubility** depends on the solute, solvent and temperature. Its temperature dependence explains crystallization and many everyday observations.

Exercise coverage: All five official sections are solved in book order: Multiple Choice, Questions for Short Answers, Constructed Response, Descriptive Questions and Investigative Questions.

A. Multiple Choice Questions

1. Matter is present in neon signs in the state of:

(a) supercritical fluid (b) plasma
(c) gas (d) liquid crystal

Answer: (b) plasma — An electric discharge ionizes the low-pressure gas in a neon sign, producing a luminous plasma.

2. Hazardous effects of shopping bags are studied in:

(a) geochemistry (b) inorganic chemistry
(c) analytical chemistry (d) environmental chemistry

Answer: (d) environmental chemistry — Environmental chemistry studies pollutants, their effects and their fate in air, water and soil.

3. The man-made polymer is:

(a) starch (b) polystyrene
(c) protein (d) cellulose

Answer: (b) polystyrene — Polystyrene is synthesized from styrene; starch, proteins and cellulose are natural polymers.

studysapiens.

4. The crystals of which substance have rhombic shape?

(a) brass
(b) sulphur
(c) graphite
(d) bronze

Answer: (b) sulphur — Rhombic sulphur is a crystalline allotrope of sulphur; brass and bronze are alloys.

5. Which liquid among the following is a colloidal solution?

(a) milk
(b) slaked lime used for white wash
(c) vinegar solution
(d) mixture of AgCl in water

Answer: (a) milk — Milk is a colloid in which tiny fat and protein particles remain dispersed. Vinegar is a true solution; the other two are suspensions.

6. Which of the following is a heterogeneous mixture?

(a) a solution of calcium hydroxide in water
(b) a solution of potassium nitrate in water
(c) hot chocolate
(d) concrete mixture

Answer: (d) concrete mixture — Concrete has visibly different, non-uniformly distributed solid components, so it is clearly heterogeneous.

7. A state of matter whose properties are between those of liquids and crystalline solids:

(a) liquid crystal
(b) supercritical fluid
(c) plasma
(d) dark matter

Answer: (a) liquid crystal — Liquid crystals flow like liquids but retain some molecular order characteristic of crystals.

8. When the tiny visible particles of a substance are dispersed through a medium, the mixture is named as:

(a) true solution
(b) colloid
(c) suspension
(d) saturated solution

Answer: (b) colloid — The intended answer is colloid: its dispersed particles are larger than those in a true solution yet remain dispersed rather than settling like a suspension.

9. A solution of $KClO_3$ has a solubility of about 13.2 g per 100 cm^3 at 40°C. How will its solubility be affected if you decrease the temperature?

(a) the solubility will increase
(b) the solubility will decrease
(c) the solubility will remain the same
(d) the solubility will first increase with temperature and then it will decrease

Answer: (b) the solubility will decrease — $KClO_3$ behaves like most solid solutes: cooling lowers its solubility, so crystals may form from a saturated solution.

studysapiens.com

10. You are studying the rate of hydrolysis of starch under different conditions of temperature. In which branch of chemistry will this topic fall?

(a) organic chemistry (b) analytical chemistry
(c) biochemistry (d) physical chemistry

Answer: (d) physical chemistry — The stated focus is reaction rate and its temperature dependence, which belongs to physical chemistry (chemical kinetics).

B. Questions for Short Answers

1.1. Why is there a need to divide Chemistry into many branches? Give three reasons.

1. Manageable study: Chemistry is vast, so branches allow its principles and substances to be studied systematically.

2. Focused expertise: Researchers can specialize, develop suitable techniques and make deeper discoveries in a defined area.

3. Useful applications: Different fields solve different problems in medicine, industry, agriculture, energy and the environment; interdisciplinary branches connect chemistry with other sciences.

1.2. Reactions may take place due to electrons present outside the nucleus or they may take place inside the nucleus. Which branches of Chemistry cover these two types of reactions?

Ordinary chemical reactions involve the rearrangement of **outer electrons** and are studied mainly in **inorganic and organic chemistry**; physical chemistry explains their rates and energy changes. Reactions and transformations **inside atomic nuclei** belong to **nuclear chemistry**.

1.3. What types of problems are solved in analytical chemistry?

Analytical chemistry separates and identifies the components of a sample (**qualitative analysis**) and determines their amounts or concentrations (**quantitative analysis**). It is used to test purity, composition and concentration in laboratories, medicines, foods, water and industrial materials.

1.4. Both graphite and graphene have hexagonal layered structures. What is the difference?

Graphene is one single two-dimensional sheet of carbon atoms arranged in hexagons. **Graphite** consists of many such sheets stacked together; strong covalent bonds act within each sheet, while weak attractions between sheets allow them to slide.

studysapiens.com

1.5. Why are supercritical fluids important?

A supercritical fluid combines liquid-like dissolving power with gas-like penetration and diffusion. Its properties can be adjusted by changing pressure and temperature, so it can act as a selective solvent and allow reactions or extractions that are difficult in ordinary solvents. Supercritical carbon dioxide is especially useful because it is non-flammable and leaves no liquid-solvent residue.

1.6. In which state does matter exist in the Sun?

Matter in the Sun exists mainly as **plasma**. At its extremely high temperature, atoms are ionized into freely moving electrons and positive ions.

1.7. What is the importance of graphene?

Graphene is exceptionally strong yet light and flexible, and it has very high electrical and thermal conductivity. Its one-atom thickness and large surface area make it important for electronics, sensors, transparent electrodes, composites and energy-storage research.

1.8. Which form of matter do most of the material things in this world belong to?

Most familiar material things and most elements under ordinary Earth conditions are **solids**. They have closely packed particles, definite shape and fixed volume.

C. Constructed Response Questions

1.1. How does a supercritical state look like?

Above the critical temperature and pressure there is no separate liquid surface or meniscus and no visible boundary between liquid and gas. The substance appears as one uniform fluid that fills the available space like a gas but can have a density and dissolving power closer to those of a liquid.

1.2. In what way is plasma created in a fluorescent tube?

A voltage across the tube accelerates a few free electrons. These electrons collide with atoms of the low-pressure mercury vapour, exciting and ionizing them; repeated collisions produce a plasma containing electrons and ions. The excited mercury emits ultraviolet radiation, and the phosphor coating converts it into visible light.

studysapiens.com

1.3. Most of the molecules we study in biochemistry are organic in nature. Where does the difference exist in organic and biochemistry branches of Chemistry?

Organic chemistry studies the structure, properties, synthesis and reactions of carbon compounds in general. **Biochemistry** studies chemicals and reactions in living systems—such as proteins, carbohydrates, lipids, enzymes and nucleic acids—and relates their behaviour to life processes. Thus, biochemistry uses organic chemistry but focuses on biological function.

1.4. Give the reason of brilliance shown by diamond. Can you improve it?

Diamond has a high refractive index, so light entering it bends strongly and can undergo repeated **total internal reflection** before leaving towards the eye. Its dispersion also separates white light into colours. Yes: precise cutting and polishing of facets at suitable angles increase internally reflected light and brilliance.

1.5. Explain the dissolution of sodium chloride in water.

1. Water is polar: its oxygen end is partially negative and its hydrogen ends are partially positive.

2. Water molecules are attracted to ions at the NaCl crystal surface. Oxygen ends surround Na^{+} ions and hydrogen ends surround Cl^{-} ions.

3. These ion–dipole attractions help overcome the ionic lattice attractions. Ions separate and become surrounded by water molecules; this is called **hydration**.

4. The hydrated ions spread uniformly through the water, producing a true solution. The ions are not destroyed or chemically changed into new molecules.

1.6. Why do different compounds have different solubilities in water at a particular temperature?

Solubility depends on the balance between attractions within the solute, attractions within water and new solute–water attractions. Compounds differ in polarity, ion charge, lattice energy, molecular size and ability to hydrate or form hydrogen bonds. At the same temperature these energy and structural factors differ, so their maximum dissolved amounts differ.

1.7. Why can NaCl not be crystallized from water just like KNO_3?

KNO_3 becomes much less soluble on cooling, so a hot saturated solution deposits many crystals when cooled. NaCl's solubility changes only slightly with temperature; cooling therefore produces relatively few crystals. NaCl is more conveniently recovered by **evaporating the water**, although cooling can still cause a small amount of crystallization.

studysapiens.com

1.8. Why is graphite slippery to touch? Which property of graphite enables it to be used as lubricant?

Graphite has layers of hexagonally bonded carbon atoms. Bonds within a layer are strong, but attractions between layers are weak, so layers slide over one another easily. This slippery layer motion and its ability to form a thin film between surfaces make graphite a useful solid lubricant.

D. Descriptive Questions

1.1. Mention the name of the branch of Chemistry in which you will study each of the following topics: (a) rate of a reaction (b) digestion of food in human body (c) properties of plasma (d) ecosystem (e) reactions taking place during fireworks (f) measurement of the absorption of wavelength with the help of ultraviolet spectrometer.

Topic	Branch of chemistry	Reason
(a) Rate of a reaction	Physical chemistry	Chemical kinetics and the effect of conditions on rate
(b) Digestion of food	Biochemistry	Enzyme-controlled chemical processes in a living body
(c) Properties of plasma	Physical chemistry	Physical state, energy and particle behaviour
(d) Ecosystem	Environmental chemistry	Chemical species and cycles in air, water, soil and organisms
(e) Firework reactions	Inorganic chemistry	Reactions and colours of metal salts and other inorganic substances
(f) UV absorption measurement	Analytical chemistry	Instrumental identification and measurement of substances

1.2. What are allotropic forms? Explain the allotropic forms of carbon and sulphur. How does coal differ from diamond?

Allotropes are different structural forms of the same element in the same physical state; the phenomenon is allotropy. Different bonding or crystal arrangements give the forms different properties.

Element	Allotrope	Structure and important properties
Carbon	Diamond	Each carbon is covalently bonded in a three-dimensional network; very hard, transparent and a non-conductor of electricity.
Carbon	Graphite	Hexagonal carbon layers with weak interlayer attractions; soft, slippery and electrically conducting along the layers.
Carbon	Buckminsterfullerene (C_{60})	A cage of carbon atoms arranged in pentagons and hexagons; discrete molecular, cage-like structure.
Sulphur	Rhombic sulphur	A crystalline form that is more stable at ordinary temperatures.
Sulphur	Monoclinic sulphur	A different needle-like crystalline arrangement, stable over a limited higher-temperature range.

Coal versus diamond: diamond is an almost pure, crystalline allotrope of carbon with a continuous covalent network. Coal is an impure, non-crystalline carbon-rich mixture containing other elements and compounds. It is black, much softer and burns readily, whereas diamond is transparent, extremely hard and does not burn easily under ordinary conditions.

1.3. What are supercritical fluids? How are they different from ordinary liquids?

A substance becomes a **supercritical fluid** when its temperature and pressure are both above its critical values. Liquid and gas phases then become indistinguishable.

Feature	Supercritical fluid	Ordinary liquid
Phase boundary	No liquid–gas boundary or meniscus	Usually has a free surface and can coexist with vapour

Feature	Supercritical fluid	Ordinary liquid
Density	Tunable; often liquid-like	Relatively fixed at a given temperature
Diffusion and penetration	Gas-like and rapid	Slower than in gases
Viscosity and surface tension	Low viscosity; no liquid surface tension at a phase boundary	Higher viscosity and measurable surface tension
Compressibility	Much more compressible; density responds strongly to pressure	Nearly incompressible
Container behaviour	Expands to fill the container	Has fixed volume and normally does not fill the whole container

1.4. Define solubility of a solute. How does the solubility of solutes change with the increase in temperature?

Solubility is the maximum amount of a solute that dissolves in 100 g of a solvent at a particular temperature to form a saturated solution.

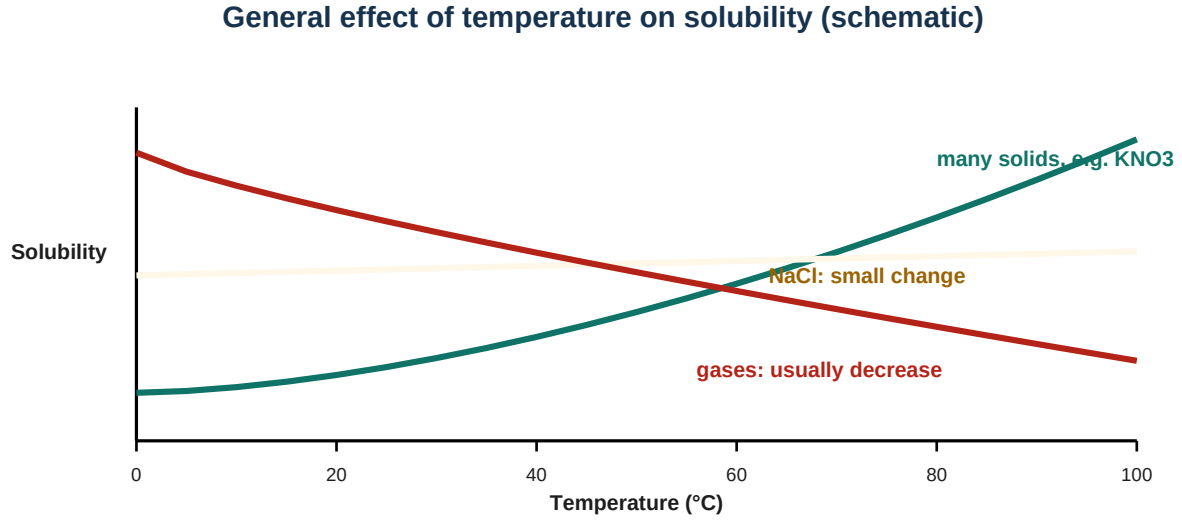
• **Most solid solutes:** solubility increases with temperature, for example KNO_3, KCl and $AgNO_3$.

• **NaCl:** solubility increases only slightly, so its curve is comparatively flat.

• **Some solids:** solubility decreases with temperature, for example $Ca(OH)_2$, Li_2CO_3 and $CaCrO_4$.

• **Gases in water:** solubility generally decreases as temperature rises; warm soda therefore loses dissolved CO_2 more readily.

Figure 1 — Original schematic of solubility trends; curves are illustrative, not data plots



Trends are substance-specific; some solid solutes become less soluble as temperature rises.

studysapiens.co

1.5. What types of movements are present in gaseous and liquid molecules?

Gas molecules move rapidly and randomly throughout the container: they show translational motion from place to place, and molecules can also rotate and vibrate. **Liquid molecules** remain close together but still slide and move randomly over short distances, while also rotating and vibrating. Unlike particles in solids, liquid molecules do not remain at fixed positions.

1.6. Differentiate between the areas which are studied under inorganic and organic chemistry.

Inorganic chemistry	Organic chemistry
Studies elements and compounds generally, including metals, non-metals, salts, acids, bases and minerals.	Studies hydrocarbons and their derivatives and other carbon-containing compounds.
Includes substances with little or no carbon, as well as simple carbon oxides, carbonates, bicarbonates and carbides.	Usually excludes the simple carbon oxides, carbonates, bicarbonates and carbides, which are treated as inorganic.
Applications include fertilizers, pigments, metals, ceramics, glass and industrial salts.	Applications include fuels, medicines, dyes, plastics, fibres, foods and biochemical substances.
Examples: NaCl, H_2SO_4, NH_3 and $CaCO_3$.	Examples: CH_4, C_2H_5OH, glucose and polystyrene.

E. Investigative Questions

1.1. Preparation of solutions leads to an important process in chemistry which enables us to purify a compound through crystallization. Describe a process in which potassium nitrate is purified by crystallizing it in water.

1. Place the impure KNO_3 in a clean beaker and add a small measured amount of distilled water.

2. Heat gently and stir. Add only enough hot water to dissolve the KNO_3; avoid a large excess, which would reduce crystal recovery.

3. Filter the hot solution through pre-warmed filter apparatus to remove insoluble impurities. Keep the filtrate hot so that premature crystallization does not block the filter.

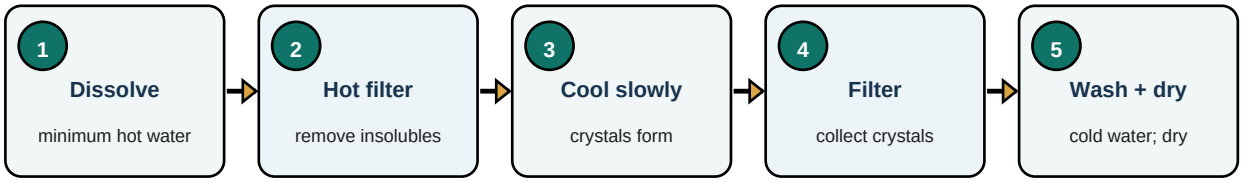
4. Allow the clear filtrate to cool slowly, first to room temperature and then, if needed, in cold water. KNO_3 solubility falls sharply, so pure crystals form while soluble impurities mainly remain in the mother liquor.

5. Filter the crystals, wash them with a very small amount of ice-cold distilled water and dry them between filter papers or in a suitable desiccator.

6. If purity is still inadequate, repeat the recrystallization. Wear eye protection and handle hot glassware and solutions carefully.

Figure 2 — Original recrystallization sequence for potassium nitrate

Purification of potassium nitrate by recrystallization



KNO3 becomes much less soluble on cooling; soluble impurities remain mainly in the mother liquor.

Safety: eye protection; use tongs or heat-resistant gloves for hot apparatus.

1.2. Graphene is called a miracle material and it is the material of the future. Which of its many properties makes it very useful in electronics?

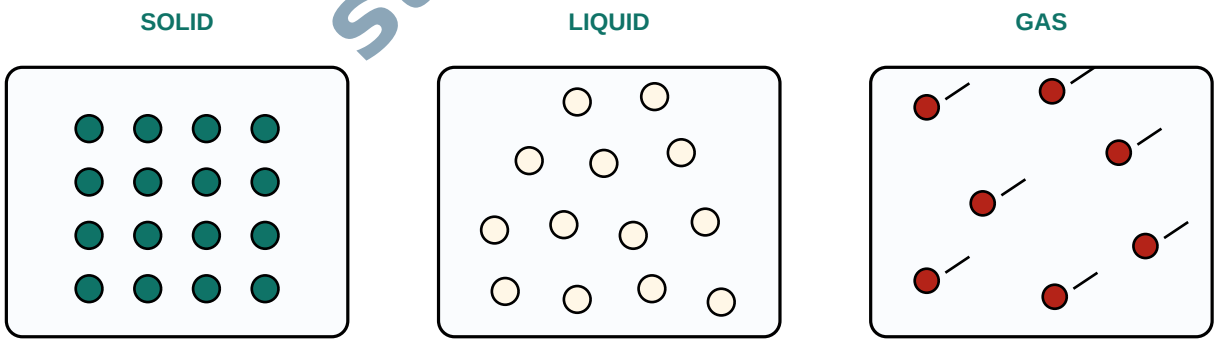
Its most important electronic properties are **very high electrical conductivity and high charge-carrier mobility**, which allow charges to move rapidly. Being one atom thick, transparent, flexible and mechanically strong also makes it attractive for transparent electrodes, flexible displays, sensors and very small high-speed devices. Its large surface area is additionally useful in supercapacitors and other energy-storage devices.

Chapter Concept Map

Matter → pure substances (elements, compounds) or mixtures → homogeneous/heterogeneous → true solutions, colloids and suspensions. Particle structure explains state and properties; solute–solvent interactions and temperature explain solubility.

Figure 3 — Original particle model of the three principal states

Particle model and changes of state



SOLID	LIQUID	GAS
closely packed	close, disordered	far apart
vibrate about fixed sites	slide past one another	rapid random motion

Heating generally increases particle energy; cooling removes energy. Intermolecular attraction also differs.

studysapiens.com

Points to Remember

Chemistry: study of matter—its composition, structure, properties, changes and associated energy.

Solid: fixed shape and volume; particles vibrate about fixed positions. **Liquid:** fixed volume but no fixed shape. **Gas:** neither fixed shape nor fixed volume and readily compressed.

Plasma is an ionized, electrically conducting mixture containing electrons and ions; it is abundant in stars.

Allotropes are structural forms of the same element with different properties, such as diamond and graphite.

Element: one kind of atom; **compound:** different elements chemically combined in a fixed ratio; **mixture:** physically combined components in variable ratio.

True solution: particles pass through filter paper and do not settle. **Colloid:** particles remain dispersed and pass through ordinary filter paper. **Suspension:** larger particles settle and can be filtered.

Unsaturated solution can dissolve more solute at the stated temperature; **saturated solution** contains the maximum dissolved amount at that temperature.

Solubility is the maximum mass of solute dissolving in 100 g solvent at a particular temperature.

Cooling favours crystallization when solubility decreases markedly with falling temperature; evaporation is more suitable for NaCl.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	b	2	d	3	b	4	b	5	a
6	d	7	a	8	b	9	b	10	d

How to attempt this chapter in the paper

- In MCQs, identify the property being tested—state, branch of chemistry, allotrope or mixture type—before selecting an option.
- For comparisons, use paired points: structure first, then the property that follows from that structure.
- In dissolution answers, name the specific attractions: Na^{+} with oxygen of water and Cl^{-} with hydrogen of water.
- In crystallization questions, connect the procedure to the solubility curve and mention hot filtration, cooling, washing and drying.

studysapiens.com